

Bharat Singh Parihar

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SUMMARY

Computer Science (Data Science) undergraduate with hands-on experience in machine learning, distributed systems, and full-stack engineering. I build production-ready AI pipelines, retrieval-augmented systems, and scalable backend architectures. Strong track record across research, hackathons, and leadership roles, with multiple SCOPUS publications and end-to-end project deployments.

EDUCATION

Symbiosis Institute of Technology B.Tech.(Hons.) CSE Core with specialization in Data Science	-Nagpur, India CGPA: 7.68/10.00 (2026)
Kendriya Vidyalaya HSC, PCM with Computer Science	- Dahi Chowki, Unnao Percentage: 87.44% (2022)

SKILLS

Languages: Python, JavaScript, TypeScript, Go, SQL, Java, C++
Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, YOLOv8, OpenCV, CNNs, RNN/LSTM, Transformers, GANs, Feature Engineering, Model Optimization, Hyperparameter Tuning, Evaluation Metrics, RAG Systems, LangChain, Vector Search (FAISS, Pinecone)
ML Systems & MLOps: Docker, Kubernetes, AWS S3/EC2/Lambda, GCP Compute, Model Deployment, CI/CD (GitHub Actions), Experiment Tracking, Data Pipelines, gRPC Services
Backend & Distributed Systems: Node.js, FastAPI, Flask, REST APIs, Microservices, Caching (Redis), Message Queues, Serverless Functions
Frontend & Full-Stack Development: React, Next.js, Tailwind CSS, ShadCN UI, Component-based Architecture, Responsive UI, State Management, Web Accessibility
Databases: PostgreSQL, MongoDB, MySQL, Neo4j, Redis, Firebase, Normalization, Query Optimization
Dev Tools & Platforms: Git/GitHub, Figma, Postman, Power BI, Streamlit, Jupyter, Linux, Docker Compose, Vite

PROFESSIONAL EXPERIENCE

PGDAV College – University of Delhi Summer Research Intern • Built a CNN-based deepfake/anomaly detection pipeline, delivering 95% accuracy and improving false-positive reduction through model tuning and data augmentation.	May 2024 – July 2024
Freelancing Website Development & Product Engineering • Designed and shipped full-stack applications for early-stage startups using React, Node.js, FastAPI, and PostgreSQL. • Implemented authentication, optimized database queries, and set up CI/CD pipelines on AWS/Firebase to reduce deployment time by 40%.	Aug 2024 – Present

PROJECTS

GenAI Realtime Assistant LangChain, LangGraph, OpenAI, Neo4j, FAISS, PyMuPDF, Pandas • Built a multi-criterial AI assistant using GraphRAG, VectorDB.	Feb 2025 – May 2025
AI-Driven Image Encryption System PyTorch, GAN, LSTM, SRDNN, NumPy, Docker • Engineered a novel GAN-LSTM encryption system for a one-pass encryption/transformation workflow.	Feb 2025 – May 2025
Federated Learning for Anomaly Detection FedML, TFF, TLS, NumPy, scikit-learn, gRPC • Addressed data privacy and bandwidth constraints as federated fraud detection pipeline with 91% accuracy without centralizing sensitive user data.	Jan 2025 – May 2025
GigX:Decentralized Freelance Marketplace Next.js, Solidity, Hardhat, Docker, Web3 • Eliminated the need for a trusted middleman in freelance payment by building a decentralized marketplace with a smart contract-based escrow and tokenized tasks on Ethereum.	Jan 2025 – Mar 2025
Medical Insurance Cost Prediction R, Shiny, glmnet, randomForest, caret, ggplot2 • Deployed a regression-based premium predictor with RMSE 300 and R ² 0.85 using Shiny dashboards.	Sep 2024 – Nov 2024
Book Detection for Visually Impaired OpenCV, PyTesseract, pyttsx3, Leptonica, ImageMagick • Built a real-time OCR-to-speech system to improve reading accessibility for the visually impaired.	Feb 2024 – Jun 2024
DeepFake Detection TensorFlow, Keras, CNN, MCDM, NumPy, Matplotlib • Created a CNN-based deepfake detector enhanced with MCDM for multi-factor prediction reliability.	May 2024 – Jul 2024
Brain Tumor Detection YOLOv8, PyTorch, OpenCV, Docker, CUDA • Deployed an MRI-based tumor detector with 89% accuracy and 15% fewer false positives.	Dec 2023 – Feb 2024

ACHIEVEMENTS

AWARDS

- 1) 🏆 **5th Place**, Hackronyx National Hackathon 2025
- 2) 🏆 **3rd Place**, Web 3.0 Hackathon, BITS Pilani 2025
- 3) 🏆 **4th Place**, CyberHack Maha Hackathon 2025
- 4) 🏆 **Winner**, GDSC Hackathon 2023
- 5) 🏆 **3rd Place**, IEEE Research Hackathon 2023

PUBLICATIONS

- [1] B. S. Parihar, et al., **Integration of Computer Vision for Book Detection and Text-to-Speech Conversion**, IEEE ICISCT 2024, Kookmin University, Korea (SCOPUS)
- [2] B. S. Parihar, et al., **The Impact of Digitalization on Psychological Treatment**, IEEE ICISCT 2024, Kookmin University, Korea. (SCOPUS)
- [3] B. S. Parihar, et al., **Realtime Cryptojacking in Advanced mobile devices**, IEEE ICPCT 2025, Amity University, Noida, India. (SCOPUS)
- [4] B. S. Parihar, et al., **Revolutionizing Industry 4.0: Multi-Level Federated Learning for Dynamic Ecosystem** – (Book Chapter) 2025
- [5] B. S. Parihar, et al., **AI and Forecasting for Renewable Energy Generation** – (Book Chapter) 2025

CERTIFICATIONS

- Fundamental of Deep Learning ~NVIDIA
- Fundamental of Accelerated Computing with Cuda C ~NVIDIA
- Cloud Data Engineer ~Google
- Accelerating CUDA C applications with Multiple GPUs ~NVIDIA
- SQL (Basic, Intermediate, Advance) ~HackerRank
- Machine Learning Specialization ~Stanford University
- Cloud Foundations and Cyber Security Foundations ~Amazon AWS
- Certified System Administrator (Global Certificate) ~ServiceNow
- Certified Application Developer (Global Certificate) ~ServiceNow

LEADERSHIP EXPERIENCE

Computer Society of India (CSI) Student Chapter

Chair

Dec 2024 – May 2025

SITNovate 24 Hours Hackathon

Organizer

19, 20 Feb 2025

Computer Society of India (CSI) Student Chapter

Founder Vice Chair

July 2024 - Nov 2024

IEEE Student Chapter

Core Member

Dec 2023 - Jun 2024

LANGUAGES

- 1) English (Fluent)
- 2) Hindi (Native)
- 3) Spanish (Basic)

REFERENCES

Prof. Dr. Nitin Rakesh

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Symbiosis International (Deemed University), Pune, India
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Prof. Dr. Geeta Aggarwal

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